Solution

Cube	**Cuboid**
Side $(a) = 12\,cm$	length $(l) = 18cm$
	breadth $(b) = 16cm$
	height $(h) = ?$

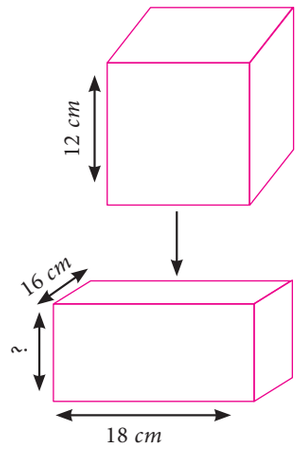


Fig. 7.27

Here, Volume of the Cuboid = Volume of the Cube

$$l \times b \times h = a^3$$

$$18 \times 16 \times h = 12 \times 12 \times 12$$

$$h = \frac{12 \times 12 \times 12}{18 \times 16}$$

$$h = 6\,cm$$

Therefore, the height of the cuboid is 6 *cm*.

Activity

Take some square sheets of paper / chart paper of given dimension 18 *cm* × 18 *cm*. Remove the squares of same sizes from each corner of the given square paper and fold up the flaps to make a open cuboidal box. Then tabulate the dimensions of each of the cuboidal boxes made. Also find the volume each time and complete the table. The side measures of corner squares that are to be removed is given in the table below.

Side of the corner square	Dimensions of boxes			Volume
	l	b	h	V
2 *cm*				
3 *cm*				
4 *cm*				
5 *cm*				

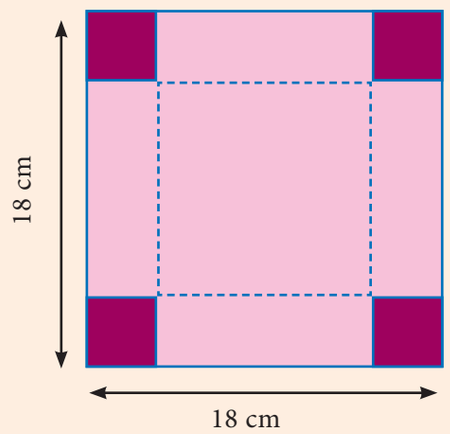


Fig. 7.28

Observe the above table and answer the following:

(i) What is the greatest possible volume?

(ii) What is the side of the square that when removed produces the greatest volume?

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